Landmark XVI-Q — Q11 (P.6 evanescent counterterm: BRST-trivial directions as sigma-images in the gauge-fixed BV variables, honest partial)

One-loop κ^3: evanescent three-point defect σ_0-exact at pole level, X non-unique (ghost-equation compatible); trivial span conditional, 43 directions unclassified, operator basis and p22 consistency open. O.1 CLOSED (scoped, κ^2 two-point) unchanged

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1 Q11: BRST-trivial directions as σ-images in the gauge-fixed BV variables

Construction

The gauge fixing is the canonical shift by $\Psi = \bar{c} \cdot F(h)$ with B integrated out. So the antifield of h enters only through $\tilde{K} = K + \bar{c} \cdot F'(h)$, and the Q08 ghost equation ($\rho = -2$) is this statement on the amplitudes. A local ghost-number -1 functional must therefore be built from $\tilde{K}$: $Y_F = \tilde{K} \cdot F(h, h)$ has the $\bar{c}$ partner $\rho\, T(K \cdot F)$, and the Q08 P family is absorbed into it. $Y_G = c^* \cdot G(c, h)$ gives X_{Khc} ($c^* \to D^T \tilde{K}$), its $\bar{c}$ partner, and the $c^* cc$ piece ($h \to s_0 h$, antisymmetrised in the two ghosts). Q08 lacked that last piece. $\sigma = (S_0, \cdot)$ is applied with the same channel operators that build the rows, with the antibracket sign -1 between antifield-type and field-type pieces. Every generator gets one test: rows, Bose and ghost equation. No per-generator fit is made.

Results

Q08 F pieces violate only the $h \cdot h \cdot c$ row. 12 generators fail through the antifield-type piece alone, because the $\bar{c}$ partner and the E channel are missing. With $\tilde{K}$: 71 of 194 nonzero F images and 8 of 33 G images lie in N, all of them also satisfying the ghost equation. The remaining failures are all in the $h \cdot h \cdot c$ row (for G, also in the ghost rows). A (λ, μ) scan, used only as a diagnostic, shows that no single normalisation works for every generator. Failures occur with and without trace monomials, so the η-trace hypothesis is not the (sole) cause. Conclusion: on the evanescent sector of the $h \cdot h \cdot c$ row, the row operators do not compose to zero with this σ. A candidate cause is the 4D-momentum rule $T(p, \cdot) = 0$ for hat components, under which $s_0 \hat{h} = 0$. This is OPEN.

Classification (conditional)

$\dim N_g = 104 = 46$ (σ span) $+ 24$ (blind) $- 9$ (overlap) $+ 43$ (unclassified). The 46 directions are trivial only if this σ is the correct linearised Slavnov–Taylor operator on the evanescent sector. No direction is certified physical. The invariant evanescent basis and the p22 κ^2 consistency check remain OPEN.

Disclosed revision

Q11-R1: the Q10 layer gate leaves code/p6q06/q07/q08/q09/q10/q11/ and tools/*q11* to the Q11 gate and treats the guard_scan3 file count as a lower bound; it is pinned by q11_expected.json. The OpenAI catalogue and the En05–En07 sheets are not premises.

Q10 layer

2 Q10: erratum on the trivial counts; anatomy of the family vectors

New manuscripts (intake, context only)

The companion source (cheungngo/enter-companion, head bb663a2, 2026-10-11) adds En05-A/B/C, En06-A/B and En07-A (partitioned Enter dynamics, operational causality, thermodynamic access) and OpenAI separation edges. The OpenAI catalogue is linked only by separations ("not a premise and not a declared path"). None of these has BRST, dimensional-regularisation, evanescent or counterterm content; nothing is imported as a premise. XV-B (modified in the same commit) stays unopened. Record: intake/new_manuscripts_q10.json.

Erratum Q10-E1 (medium)

Withdrawn: the Q08 counts "≥ 97 in the trivial span at the best fitted λ", "≥ 55 trivial" and "≥ 36 unexplained" as a count of non-trivial directions; the Q09 counts "F verified on 139/204" and "derived trivial span 59 (27 in the ghost-equation subspace)" as counts of BRST-trivial directions. Reason (Q10_family_anatomy): of the 204 F generators, 15 vanish after the h-leg symmetrisation; all 124 nonzero generators that pass have *each piece separately* in N, so the λ test is vacuous on them, and 62 of them are supported only on blind coordinates. 64 of the 65 failures admit no λ at all. For G, 28/33 admit no λ (5 need $i/2$); for P, 132/134 admit no λ (2 need -2). The Q08 hand-built pieces are not σ-images of the stated Y for most generators. The defect lies in the construction, not the normalisation, so no normalisation (derived $-1, -1, +1$ or fitted $-1, i/2, -2$) can repair it.

Blind directions and the restated bound

42 X coordinates (30 in the $K{\cdot}c{\cdot}h$ sector, 12 in the c^*cc sector) enter no three-point row and no Bose condition. 24 of these survive the ghost equation. They are free at this order and unconstrained by the rows. $\dim N - \dim(N \cap$ span of all Q08 pieces$) = 152 - 116 = 36$ is exact linear algebra, but the spanning pieces are not verified σ-images. So 36 bounds only the directions outside that span. It is neither a lower bound on physical directions nor an upper bound on trivial ones.

Open

The F trace-type failures (the η-trace hypothesis is neither confirmed nor excluded); a correct construction of $\sigma(Y)$ for $Y = K{\cdot}F$, $c^*{\cdot}G$, $\bar{c}{\cdot}\Phi$ (or a proof of absence); classification of N; the invariant evanescent operator basis and X in it; consistency with the p22 κ^2 evanescent counterterms.

Disclosed revision

Q10-R1: the Q09 layer gate leaves code/p6q06/q07/q08/q09/q10/ and tools/*q10* to the Q10 gate; it treats the guard_scan3 file count as a lower bound (Q10 files are scanned, not exempted); it is pinned by q10_expected.json. The Q05–Q08 gates are unchanged.

Q09 layer

3 Q09: review fixes; derived BRST-trivial directions of X

Q07 review fixes

(i) Status relabelled: *evanescent κ^3 three-point defect σ_0-exact at pole level; X non-unique (ghost-equation compatible).* This supersedes the Q07/Q08 wording "σ_0 closure CLOSED at pole level"; the frozen Q07/Q08 certificates are left unchanged. (ii) The Q04 independent rebuild report is bound by its sha256 in the Q09 gate. (iii) A new static scan module (guard_scan3) flags operator.methodcaller in any form and coefficient/exponent walks aimed at d (coeff, as_coeff_exponent, as_independent, collect, as_poly, degree, and dictionary walks of the power/coefficient dictionaries keyed by d). All 10 probes are flagged; code/, tools/ and tests/ are clean. The Q05 scan is unchanged; the runtime guard stays primary.

(a) Derived normalisations

For $Y = K \cdot F(h,h)$ the components of $\sigma Y = (S_0, Y)$ are produced by the same operators that build the rows (X_{hhh} by the $\delta S_0/\delta h$ contraction, X_{Khc} by $h \to s_0 h(c)$). These operators commute, so $\sigma^2 = 0$ fixes the relative sign $\lambda_F = -1$; likewise $\lambda_P = -1$ for $Y = \bar{c} \cdot \Phi$, and $\lambda_G = +1$ for $Y = c^* \cdot G$. Check: at $\lambda_F = -1$, 139 of 204 generators lie in the kernel. All 65 failures carry a K- or h-trace monomial, which points to the η-trace convention in the probe contraction (open). The Q08 G and P constructions do not close at the derived values (0/33, 0/134); they were assembled by hand and are open. The Q08 fitted counts are superseded.

(b)–(d)

(b) Derived trivial span: 59 directions, 27 of them inside the 104-dimensional ghost-equation subspace. The rest are not classified. (c) No invariant evanescent operator basis was constructed: open. (d) Consistency with the p22 κ^2 evanescent counterterms: open.

Disclosed revision

Q09-R1: the Q08 gate leaves code/p6q06/q07/q08/q09/ and tools/*q09* to the Q09 gate; it is pinned by the Q09 expected-values file. The Q05–Q07 gates and guard_scan2 are unchanged.

Q08 layer

4 Q08: structure of the κ^3 evanescent counterterm

Scope

Pole level, one loop, κ^3, three-point rows (c$\phi\phi$, cc$\bar{c}$, ccK, hhc) as in Q07. Unknowns: the local evanescent counterterm X of Q07 (821 coordinates: X_{Kch} 134, $X_{\bar{c}ch}$ 81, X_{c^*cc} 18, Bose-symmetric X_{hhh} 588).

1. Ghost equation

The loop poles obey $\Gamma_{\bar{c}ch} = \rho\,\Gamma_{Kch}[K \to T(\bar{c})]$, $T(\bar{c}) = \mathrm{sym}(k, \bar{c}) - \frac{1}{2}(k \cdot \bar{c})\eta$, with one constant $\rho = -2$ (graph by graph: G04/G03 and G09/G07). Imposing $X_{\bar{c}ch} = -2X_{Kch}[K \to T]$ keeps the system solvable; a particular solution is certified ($Ax = b$ verified). Free directions drop from 152 to 104. The η-trace is taken at the $d \to 4$ residue level (hat trace 4, tilde trace 0).

2–3. Free directions and trivial families

Kernel N of the rows: dim 152 (104 with the ghost equation). Single-sector blind directions: 69 in X_{Kch} alone, 12 in X_{c^*cc} alone, 26 in X_{hhh} alone (σ_0-closed evanescent cubic structures, candidates for cubic parts of invariant evanescent operators), none in $X_{\bar{c}ch}$ alone. BRST-trivial candidates $\sigma(Y)$, gh -1: $Y = K \cdot F(h, h)$ (field redefinition of h), $Y = c^* \cdot G(c, h)$, $Y = \bar{c} \cdot \Phi(h, h)$ (gauge fermion). The relative normalisation of the two pieces of each family was fitted (best uniform λ: -1, $i/2$, -2); with it at least 97 directions of N (55 inside the ghost-equation subspace) are trivial, at most 116. At least 36 directions are not shown trivial. Whether they are physical ambiguities (invariant evanescent operators) is not settled, and no explicit operator basis was constructed.

4. Consistency with p22

Open: the cubic completion of the p22 κ^2 evanescent operators was not built.

Outcome

P.6 Partial. Q07's pole-level σ_0 closure on the three-point rows now holds with a ghost-equation compatible X. Not implied: four-point rows, finite parts, uniqueness/physicality of X, p22 consistency.

Disclosed revision of a frozen layer

Q08-R1: the Q07 gate (code/p6q06/q07/release_q07.py) leaves files under code/p6q06/q07/q08/ and tools/*q08* to the Q08 gate; it is pinned by the Q08 expected-values file. The Q05 and Q06 gates are unchanged.

Q07 layer

5 Q07: κ^3 three-point rows — h h c, evanescent counterterms, pole-level σ_0 closure

Scope

One loop, κ^3, UV-pole level, declared XVI-P prescription (DRED, opaque d, MS-bar, Feynman $+i0$). Rows of $(S, \Gamma_1^{\text{pole}})$ with three external legs: $c\phi\phi$ (closed, Q05), $cc\bar{c}$ and ccK (4D part closed, Q06), and hhc (this release). Four-point and higher κ^3 rows and finite parts are not addressed.

Row h h c

Channels: (A) Γ_{hhh} (G01+G02+G08+G14+G15) with one leg replaced by $s_0 h(c)$; (B) own Γ_{hh} pole (graviton, ghost and scalar loops) with one leg $s_1 h = \frac{1}{2} L_c h$; (C) $\delta S_{EH,3}/\delta h \cdot \Gamma_{Kc}$; (D) $\delta S_2/\delta h$ (EH + gauge fixing) $\cdot \Gamma_{Khc}$; (E) $s\bar{c} = \sigma_F F(h)$: $F \cdot \Gamma_{\bar{c}ch}$ (the channel missing from the Q04 ledger, erratum Q06-E2). The tree row $(S,S)/2|_{hhc}$ closes exactly with $\sigma_F = i$ and requires channel E. The scalar loops G14, G15 and the scalar part of Γ_{hh} were recomputed with the Q05 engine; the matter sector alone closes exactly (incl. evanescent terms). Result: the 4D part of the row vanishes identically with all channels at $+1$ (the unconstrained fit has the unique solution all-equal), every single-channel deletion and a flip of σ_F re-open it; an explicit evanescent remainder (248 terms, every term containing a $\tilde{g}$ contraction) is left.

Evanescent counterterms

(i) The naive p22-type choice (subtract the $\tilde{g}$ part of every loop input) is *not* σ_0-consistent: tree vertices regenerate $\tilde{g}$ structures from 4D loop parts (22 / 26 terms left on ccK / $cc\bar{c}$). (ii) Joint test: $CT = -\Gamma^{\text{pole}} + X$ with X a generic local, purely evanescent κ^3 functional (monomials with at least one $\tilde{g}$) for Γ_{Kch}, $\Gamma_{\bar{c}ch}$, Γ_{c^*cc} and a Bose-symmetric Γ_{hhh}; the same X_{Kch}, $X_{\bar{c}ch}$ enter the ghost rows and hhc. The linear system over $\mathbb{Q}(i)$ (821 unknowns) is solvable; a particular solution ($X_{c^*cc} = 0$) is certified and verified by substitution; 152 parameters stay free (non-unique). Not tested: ghost-equation compatibility of X, four-point rows.

Outcome

σ_0 closure at κ^3: CLOSED at pole level on the one-loop three-point rows in the enlarged (evanescent) counterterm domain. P.6 remains Partial: CT-domain reduction to an invariant operator basis, four-point rows, finite parts, ghost-equation compatibility and uniqueness of X are open. O.1 (scoped, κ^2 two-point) unchanged; bulk symmetry restoration not claimed.

Disclosed revision of a frozen layer

Q07-R1: the Q06 gate (code/p6q06/release_q06.py) leaves files under code/p6q06/q07/ and tools/*q07* to the Q07 gate and treats the pinned Q05 guard-scan file count as a lower bound (as Q06-R1 did for the Q05 gate). The Q05 gate is byte-identical to Q06. The revised Q06 gate is pinned by the Q07 expected-values file.

Q06 layer

6 Q06: P.6 continued (honest partial)

Scope and standing constraints

One-loop κ^3 three-point sector at UV-pole level, XVI-P declared prescription (DRED, opaque d via qguard, MS-bar, Feynman $+i0$), own Feynman rules of Q05. XVI-O/N/P read-only; O.1 remains CLOSED (scoped, κ^2 two-point) and is not affected; bulk symmetry restoration is not claimed; insertion frozen; no Missing is ever set to zero.

1. Ghost rows $c\,c\,\bar{c}$ and $c\,c\,K$: diagnosis and fix

Q05 reported both rows OPEN: the assembled one-loop row did not vanish and no rescaling of the four channels closed it. Q06 splits every channel by orientation of the two identical ghosts and fits all six coefficients freely on the 4D (non-evanescent) part. The fit has a unique solution (up to scale), every coefficient ± 1: the h channel and both c channels enter as assembled in Q05, but the K channel $\delta S_1/\delta h|_{Xc_j} \cdot \Gamma_{Kc_i}$ enters with the *opposite* orientation sign. Reason: in the K channel the external c_i sits inside Γ_{Kc_i}, to the right of c_j, while in the h channel the order is reversed; bringing the monomial to the row ordering $c_1 c_2$ costs one Grassmann transposition. The tree row has no K channel, so the tree calibration ($\sigma_c = -1$ for $\bar{c}$, $+1$ for K) is unchanged. With this fix the 4D part of both rows vanishes exactly, and every single-channel mutation (drop $h_{1,2}$, $K_{1,2}$, c_1, c_2, flip σ_c, restore the Q05 K sign) leaves a nonzero 4D residual.

Second route for G10. The $c^* cc$ triangle was recomputed with a re-routed loop momentum (graviton carries $l - p_1$) and the two c slots of V_{c^*cc} entered in the opposite order (Grassmann sign -1): bit-identical to the Q05 pole. G10 was therefore not the defect.

What remains. A purely evanescent remainder (22 terms for ccK, 24 for $cc\bar{c}$; every term carries one T between external polarisations). Because the row is linear and its 4D part vanishes, local evanescent counterterms equal to minus the evanescent pole parts (p22 prescription) remove it; whether those counterterms are s_0-consistent on the full κ^3 evanescent domain is OPEN; it is recorded explicitly (sha256 in the certificate) and is not called an obstruction.

2. hhh graphs G01 and G02

A sparse-ring fast path (same algorithm as the Expr path, partial products truncated in the Taylor parameter at the pole order) reproduces G05 and G06 bit-identically and makes G01 (V_3^3 triangle) and G02 feasible. G02 needs the quartic vertex: V_4 was derived from S_{EH} by expanding $\sqrt{-g}\, g^{mn} g^{ac} g^{bd}$ to second order. No non-scaleless XVI-P graph contains V_4 (its tadpole is scaleless), so V_4 is validated by the exact Einstein–Hilbert gauge identity $\delta_0 S_4 + \delta_1 S_3 = 0$ ($\delta_0 h = \partial\xi + \partial\xi$, $\delta_1 h = L_\xi h$), which vanishes identically; the same identity at $n = 2$ (V_3 vs V_2) is the machinery control. G02 (three channels) and G01 are Bose-symmetric under $(a_1, p_1) \leftrightarrow (a_2, p_2)$, and G02's channels map onto each other. All 15 P.6 graphs now have pole values (G14/G15 from Q04).

3. Row $h\,h\,c$, κ^3 defect, counterterms, σ_0 closure

Not assembled: Missing. The channel ledger is recorded, including a channel absent from the Q04 ledger: with b eliminated, $s\bar{c} \propto F(h)$, so $F(h) \cdot \Gamma_{\bar{c}ch}$ (G04+G09) enters (erratum Q06-E2). The κ^3 counterterm-domain reduction, defect, counterterm solve and σ_0 closure are not established.

Errata

Q06-E1 (low): Q05's ghost rows were OPEN because of the K-channel Grassmann sign in the assembly, not because of the graphs; corrected here, Q05 record kept. Q06-E2 (low): Q04 h h c ledger lacks the $F(h) \cdot \Gamma_{\bar{c}ch}$ channel.

Disclosed revision of a frozen layer

Q06-R1: the Q05 gate (code/p6q05/release_q05.py) now treats the pinned Q05 guard-scan file count as a lower bound (the recursive scan grows when Q06 adds code) and leaves Q06-owned files to the Q06 gate; Q05_guard_scan.json was regenerated (26 to 33 files, still clean). The revised Q05 gate is re-pinned by the Q06 expected-values file.

Status

P.6 Partial. execution_pass true; claim_status partial; scope: pole level, one loop, κ^3. Next: evanescent s_0-exactness of the ghost-row remainder; h h c row; κ^3 counterterm domain, defect, counterterm solve and σ_0 closure.

Q05 layer

7 Q05: P.6 continued (one-loop κ^3 three-point sector) — honest partial

Scope. Pole level, one loop, κ^3, under the XVI-P declared prescription (DRED with opaque d guarded by `qguard`, MS-bar, Feynman $+i0$, evanescent structures kept as tilde metrics). O.1 stays CLOSED (scoped, κ^2 two-point) and is not changed; `bulk_symmetry_restored` stays false.

Own gravity and ghost Feynman rules

Derived in `code/p6q05/gravity.py` from the XVI-P declared action: $S_{EH} = -(2/\kappa^2)\int L_{\Gamma\Gamma}$, the gauge fixing obtained by eliminating b ($+D_\mu D^\mu$), $S_{gh} = \int i\bar{c}_\mu D^\mu[(\partial c + \partial c) + \kappa L_c h]$, $K\,sh$ with $sh = \frac{1}{2}L_c(\eta + \kappa h)$, and $sc = \frac{\kappa}{2}c\cdot\partial c$, $s\phi = \frac{\kappa}{2}c\cdot\partial\phi$ (XVI-P P13 s-images). Checks: the EH quadratic form is gauge invariant. EH plus gauge fixing equals $k^2(I - \frac{1}{2}J)$ (XVI-P $K = -\Box P$), so the graviton propagator is $i/k^2\,(I - \frac{1}{2}J)$ with $\mathrm{tr}\,\eta = 4$ (Q4S). Validation against pinned XVI-P two-point poles, all structures including evanescent ones, constant ratio (own/XVI-P): D01 (V3×V3) $= I$, D03 (ghost loop) $= I$, D06 $(\bar{c}c) = I$, D13 $(K'\cdot c) = -I$. I is the Q04 convention factor (D11). **Disclosure:** D13 has the opposite overall sign: the orientation of the $K'\cdot c$ slot differs between the two codes. Own rows use own D13 consistently. The ghost equation holds at two points ($D06 = -2\,D13[K \to T]$) and graph by graph at three points ($G04 = -2\,G03[K \to T]$, $G09 = -2\,G07[K \to T]$).

Graphs (UV pole residues, own rules)

Evaluated: G03–G13; nonzero poles are certified in `Q05_P6_graphs.json`. Q04 carried G14, G15 and the scaleless T1–T3. **Missing** (never set to zero): G01 ($V3^3$, implemented, evaluation did not finish within the release budget) and G02 (needs V4, not constructed).

Identity rows at the pole

Antifield-channel signs are calibrated on the classical $(S_0, S_1) = 0$ (tree level) and used unchanged at one loop. **Row $c\phi\phi$: closed.** The residual is identically 0. It needs the channel $\delta S_1/\delta h|_{\phi\phi}\cdot\Gamma_{Kc}$, which the Q04 ledger omitted (erratum Q05-E1). Detected mutations: flipped σ, dropped K-channel, dropped G06/G12/G13. G05's s_0-projection vanishes identically, so dropping G05 is undetectable in this row; that is recorded, not a failure. **Rows $cc\bar{c}$, ccK: OPEN.** The tree identities close with one constant σ_c. The one-loop rows do not close, and no rescaling of the four channels closes them. They are *not* reported as an obstruction. Since the ghost equation holds, the defect is located in G10 (one sign bug already fixed, erratum Q05-E4), in the K/c channels, or in the Grassmann bookkeeping of the two identical c legs. **Row hhc: Missing** (needs G01, G02).

Counterterms and σ_0 closure

In the $h\phi\phi$ and $\phi^* c\phi$ sector, the MS-bar counterterm is minus the certified poles. Its consistency in row $c\phi\phi$ is exactly the closed row. Every other κ^3 sector is Missing or Open. The κ^3 counterterm-domain reduction is not done. **σ_0 closure at κ^3 is not established.** Finite parts are Missing.

Q04 review folded in

The static scan is hardened (`guard_scan2`: recursive over `code/`, flags all 7 reviewer evasion forms). The claim is downgraded to a *best-effort policy scan, not a security boundary.* The Q05 gate rejects any unpinned file in `code/**`, `tools/` or `tests/`. The `scalar_loop3.py` docstring (G13/G14 should read G14/G15) is erratum Q05-E2; the file is pinned by the Q04 layer gate and left unchanged. Pinned Q04 tools are left unchanged, and Q05 adds `build_q05.py`, `source_set_q05.py`, `rebuild_q05.py` and `make_expected_q05.py`.

Next step

Independently check G10 and the c-channel, then close or diagnose the ghost rows. Evaluate G01 (lower-cost contraction) and construct V4 for G02, which gives row hhc. Then reduce the κ^3 counterterm domain and attempt σ_0 closure.

Q04 layer

8 Q04: P.6 entry — one-loop κ^3 three-point sector (honest partial)

P.6 is the one-loop κ^3 three-point sector. It lies outside the O.1 scope (one-loop κ^2 two-point), and **O.1 CLOSED (scoped) is unchanged**. Q04 releases the stages listed below. Anything not evaluated is marked *Missing*, never set to zero.

8.1 Inventory and vertices (Constructed)

An independent enumerator (one-loop 1PI graphs as cycles of at most 3 vertices; each vertex of an $E = 3$ graph carries at least one external leg) is run on the p14r loop-capable vertices plus the κ^3 vertices. It reproduces XVI-P P23's 15 loop graphs exactly (hhh 5, h$\phi\phi$ 4, $\bar{c}ch$ 2, Kch 2, ccc^* 1, $\phi^* c\phi$ 1). It adds three κ^3 tadpoles: V5_grav $\to$ hhh, and V_hhh$\phi\phi$ with an hh line $\to$ h$\phi\phi$ or a $\phi\phi$ line $\to$ hhh. All three are scaleless (massless self-contraction; $\int d^d l\, l^{2k}/l^2 = 0$, no log member), so they are derived 0.

Power counting: at one loop $I = V$, so $\sum_v(\text{legs}_v - 2) = E$ and the κ power equals E (checked on every enumerated graph). Hence the one-loop κ^3 one-point and two-point sectors are empty.

κ^3 vertices: the ghost, antifield and gauge-fixing vertices are exact at their order. $sh = \frac{1}{2}L_c(\eta + \kappa h)$ is affine in h, Ψ is linear in h, and sc, $s\phi$ are exact at κ. The only new vertices are therefore V5_grav and V_hhh$\phi\phi$. The cubic term of $\sqrt{-g}g^{\mu\nu}$ is derived in closed form and agrees with an independent exact determinant/inverse series; a mutation is detected. V5_grav is not constructed, because it enters only the scaleless tadpole.

8.2 Identity at κ^3 (ledger Constructed)

$(S, \Gamma_1) = A_1$ at κ^3 is $(S_0, \Gamma_1^{[3]}) + (S_1, \Gamma_1^{[2]}) + (S_2, \Gamma_1^{[1]}) + (S_3, \Gamma_1^{[0]})$. The last two terms vanish ($\Gamma_1^{[1]} = 0$ by D14; $\Gamma_1^{[0]}$ is field independent). The ghost-number-1 three-field rows are enumerated mechanically: 14 rows. Four are odd under ϕ-parity (zero), six contain auxiliary or nonminimal fields, and four need P.6 inputs (chh, $c\phi\phi$, $cc\bar{c}$, ccK); each needed input is recorded. The κ^3 counterterm domain is given only as structural monomial counts (4D versus evanescent); its BRST/σ_0 reduction is Missing.

8.3 Evaluation (Partial)

Own UV-pole engine (`code/q04/uvpole.py`). It uses its own Feynman rules, symbolic opaque d through qguard, and the DRED hat/tilde split (QdS momenta, Q4S vertex indices); MS-bar residues are taken only through `dred.substitute_d`. The UV pole is extracted by an exact Taylor (IR-rearranged) expansion. *Validation:* the engine's scalar bubble equals $i\times$ XVI-P's P18 D11 pole on all 14 structures, including the 8 evanescent ones. A vertex mutation breaks the agreement.

Evaluated: P6-G14 (scalar triangle) and P6-G15 (scalar bubble), the scalar-loop contributions to hhh, at the UV-pole level (exact residue; Bose symmetric). **Matter Ward identity at three points**: diffeomorphism invariance of the scalar-loop $W[g]$, i.e. the matter part of row chh, with Γ_2 the validated D11. The pole residue satisfies it exactly, including the evanescent structures. A wrong Lie-derivative sign and dropping the $\sqrt{-g}g^{-1}$ quadratic-vertex bubbles are both detected. The d-dependent coefficient differs from zero by $O(d-4)\times U$, which is finite and not evaluated.

Missing: 13 graphs, namely every graph with a graviton or ghost loop, V3/V4_grav, the ghost/antifield vertices or mixed lines (P6-G01..G13). Also Missing: their finite parts, the κ^3 defect assembly, the counterterm solve and σ_0 closure.

8.4 Next step

Implement V3_grav (own rules, validated against XVI-P D01) and the ghost vertex (validated against D06/D13). Evaluate G01–G13 at the pole level, starting with the rows $c\phi\phi$ (G05, G06, G11, G12, G13) and $cc\bar{c}/ccK$ (G03, G04, G07, G09, G10). Assemble the projected κ^3 defect, reduce the counterterm domain, solve, and check σ_0 closure.

8.5 Q03 review findings (low), folded in

(i) `tools/source_set.py`, which decides which files count as source, is now pinned in the gate together with `build.py` and `rebuild_from_source.py`; the gate also re-hashes the Q03 independent report. (ii) The d guard cannot see code that leaves sympy symbolics. This covers a pure-Python term walk that identifies d by name (comparing with `== d` is already refused by the p22 guard), a `str`/`srepr` rename followed by a re-parse, and raw dense-list evaluation. The probes in `certificates/Q04_guard_scan.json` show these *do* bypass the runtime guard, and they cannot be closed at runtime. The class is closed by policy: an AST static scan of every package file (`code/q0*/*.py`; the guard probe files are exempt) forbids the escape primitives, and the gate requires the scan to be clean.

Q03 layer

9 Q03: correction release (no new science)

Q03 corrects the O.1 closure record after the Q02 independent review (report `reproducibility/independent_rebuild_report_q`
sha256 `e4b54702bc76...`, reviewed source zip `378651a3...`, tree `4b551b98...`; decision: CONFIRM CLOSED (scoped)). The Q02 certificates are a frozen layer; the authoritative record is `certificates/Q03_O1_record.json`.

9.1 O.1 status

`O1_closure` = CLOSED (scoped), `O1_closure_confirmed` = true, bound to the report sha256 and the reviewed tree (the gate re-hashes the report). Scope (verbatim from the review): one-loop, order κ^2, bulk two-point retained sector of $S_{\rm inv}$ ($h, \phi, c, \bar{c}, b$ + antifields; gh0 seeds), $m = 0$, frozen gauge fermion, under XVI-P's declared prescription (DRED: QdS loop momenta, Q4S vertex indices, MS-bar) including the p22 evanescent counterterm contract revision.

9.2 Correction 1: counterterm domain of O.1 item 5 (medium)

The counterterm domain used is the *enlarged* domain: the 11 p14r 4D directions over $\mathbb{Q}(i)$ *plus* the 10 p22 evanescent operators ($K' \cdot c$ E2, coefficient $-1/16$; $\bar{c}c$ SE, $1/8$; eight $h \cdot h$ evanescent structures), 21 directions in all. With QdS (d-dimensional) external momenta the local divergent part contains evanescent poles that only the p22 operators cancel. "ST4a in the declared p14r domain" on its own holds only for 4D externals. σ_0 closure including the evanescent operators was verified in the p20/p22 independent reviews.

9.3 Correction 2: prescription; XVI-O closure reason 8 (medium)

The closure uses XVI-P's prescription, not XVI-O's. XVI-O's o70-sync D11 matter-loop kernel counterterm (closure reason 8) is superseded on evanescent legs: both XVI-O kernel continuations fail there (p22). Reason 8 is therefore recorded as NOT ANSWERED in XVI-O's terms; reasons 1–7 are answered as in Q02, and reason 7 now names the enlarged domain.

9.4 What O.1 CLOSED (scoped) does not imply

Not the functional/all-n-point ST4b; not bulk symmetry restoration (`bulk_symmetry_restored` = false); no change to XVI-O's own ledger (read-only, O.1 there stays NOT_CLOSED); not the three-point sector / P.6 (κ^3) or higher orders; not the insertion (frozen); not finite amplitudes with ϵ-scalar externals.

9.5 Caveats

(i) Both independent reviews ran on the same box as the author (shared Python/sympy and TeX Live). (ii) The D01/D03 pole values were never recomputed from independent Feynman rules; they are cross-checked only through the exact cancellation of the $c \cdot h$ defect in d. (iii) Full σ_0 closure and the $c \cdot h$ defect rely on the earlier p19–p22 independent checks. (iv) D06 is accepted through its exact equality with XVI-P P15 Σ.

9.6 Correction 3: gate and guard (low)

The gate now re-hashes every pinned provenance file (all 33, and it rejects any unpinned provenance file), every pinned code/conventions file and the bound independent report at gate time. It also checks that every file it reads is in the source set or is a build output. The d guard now refuses value assignment (`evaluate`/`subs`/`__call__`/`compose`) on `PolyRing`/`FracField` elements whose generators include d when called directly. All 8 reviewer forms raise, and the 18 earlier forms still raise. Disclosed residual: raw dense-list evaluation carries no symbol information and is not guarded; no package code uses it.

Q02 layer

10 Q02 scope and outcome

Q02 evaluates the bulk O.5 graphs D04, D05a and D06, reconciles D12, states XVI-O's ST4a/ST4b on the full one-loop κ^2 bulk two-point sector of $S_{\rm inv}$ ($m = 0$), solves the bulk counterterms, checks full σ_0 closure, and walks the O.1 checklist. XVI-O, XVI-N and XVI-P are read-only and hash-pinned. The prescription is XVI-P's declared one: DRED with QdS loop momenta and Q4S vertex indices; d opaque behind the XVI-Q guard; MS-bar; Feynman $+i0$; the p22 evanescent counterterms. Normalisation: $i\Gamma_1 = \sum$ graphs. Coefficients are given per κ^2, in units of $M_B = \hat{I}(1,1) = 1/\epsilon + 2 - L + O(\epsilon)$, with $L = \log(-p^2/\mu^2 - i0)$.

11 Bulk graphs

graph	content	value	evidence
D04	$\phi\phi$, two $V_{h\phi\phi}$, h+ϕ lines	0 exactly in d	two routes + hand contraction
D05a	$\phi\phi$, $hh\phi\phi$, h loop	0 (derived)	numerator l-independent, $\omega = 2$, no log r
D06	$\bar{c}c$, two $V_{\bar{c}ch}$, h+c lines	$\delta^\rho_\nu p^4$: $-\frac{4d-7}{8(d-1)}$; $p^\rho p_\nu p^2$: $\frac{3(d-2)}{8(d-1)}$; $\tilde{G}^\rho{}_\nu p^4$: $-\frac{3}{8(d-1)}$	= XVI-P P15 Σ exactly
D12	hh, $hh\phi\phi$, ϕ loop	0 (derived)	$\omega \in \{2, 4\}$, no log member

D04: mechanism. The D04 bubble at $p \neq 0$ is *not* scaleless, so its zero needs an explicit reason.

Take the vertex $V_{ab}(p,q) = p_a q_b + p_b q_a - \eta_{ab}\, p\cdot q$ with $q = l - p$, and the graviton numerator $P = \frac{1}{2}(\eta\eta + \eta\eta) - c\,\eta\eta$, where the vertex-index trace is $\eta \cdot \eta = n$. Then

$$V \cdot P \cdot V' = 2p^2q^2 + (n-2)(p\cdot q)^2\,[1 - c(n-2)].$$

The $(p\cdot q)^2$ term is the coupling to the trace mode. It cancels exactly when $c = 1/(n-2)$, which holds for the de Donder propagator in DRED ($n = 4$, $c = \frac{1}{2}$, the declared prescription) and in CDR ($n = d$, $c = 1/(d-2)$). What remains is $2p^2(l-p)^2$. This cancels the scalar propagator and leaves $p^2 \int 1/l^2$, which is scaleless with $\omega = 2$ and no logarithmic member, so D04 = 0 exactly in d.

D04 = 0 is therefore a statement about de Donder gauge, minimal coupling and $m = 0$, off shell. It is not a general property of the bubble, and no free-EOM equivalence is used.

D04: routes.

- Engine route 1: Feynman parameters, with no scaleless rule.
- Engine route 2: an invariant fit identical to $\frac{1}{2}p^2(l-p)^2$, followed by IBP.
- Route 4: fully independent of the engine. It uses its own covariant algebra and its own Feynman-parameter integration with symbolic D for arbitrary propagator powers.

In route 4, both the DRED and CDR cases give 0. The mutations, as ratios to the master, are:

- mixed scheme ($n = 4$, $c = 1/(D-2)$): $\frac{D-4}{2(D-2)}p^4$, i.e. $O(\epsilon)\times$pole;
- vertex trace normalisation $\frac{1}{2}$: $p^4/4$;
- non-de Donder gauge pieces of Fröb's propagator: $T^{(4)}$ gives $(D-1)p^4$ and $T^{(5)}$ gives $\frac{(D-2)(D-1)}{8}p^4$;
- non-minimal coupling ξ: $-\xi p^4$, which is linear in ξ;
- engine mutation dropping the trace term of the propagator: $-i/8$.

The gauge piece $T^{(3)}$ gives 0, and this is expected rather than a failure: $l^a V_{ab} = p_b q^2 + q_b p^2$, so every $T^{(3)}$ term carries q^2 and is scaleless. It is recorded as a Ward-identity zero, not counted as a detector.

D04: literature. M. B. Fröb, CQG 35 (2018) 055006 (arXiv:1710.00839), works in flat space, in a general gauge, in n dimensions. The de Donder part of the graviton-loop scalar self-energy (his eq. inv2pf_termsumint_grav) has an integrand bracket proportional to m^2 or ξ, and "vanishes identically" for $\xi = 0$, $m = 0$, in accordance with Kahya and Woodard (2007). For $m = 0$ his ξ^2 terms vanish, and the gauge-parameter pieces contribute. **All three statements agree** with our results. The 1PI self-mass is gauge dependent; D04 = 0 holds in the declared gauge only.

E7. The symmetry factor of D04 is 1. XVI-O's "$\frac{1}{2}$" was marked Unverified and never used for a value. XVI-P never evaluated D04, and its D11, which has two identical ϕ lines, keeps $\frac{1}{2}$. No earlier value changes.

D05a, D12. These are genuine one-propagator tadpoles: no external momentum flows through the loop line. They are pure power divergences ($\omega = 2$, and $2, 4$ for D12) with no logarithmic member, so there is no pole on either side, and the zero is not a cancellation between UV and IR poles. Fröb also notes that graviton tadpoles vanish in dimensional regularisation around flat space.

D06. This is the full $\bar{c}c$ two-point function. MS-bar poles: $-3/8$, $1/4$, $-1/8$. Local finite parts: $-2/3$, $5/12$, $-1/3$. Coefficients of L: $3/8$, $-1/4$, $1/8$. It is identical in d to the ghost self-energy Σ of XVI-P P15, for which the ghost equation against D13 is verified.

D12 reconciliation. XVI-O never labels D12 "Missing". The word survives only as stale p14r text in XVI-P's P23 O.5 line. XVI-P p15 derived D12 = 0, and Q02 confirms this independently. The value is derived, never set by convention; this is erratum E6.

12 Bulk retained identity, counterterms, closure

XVI-O's definitions, quoted verbatim:

- ST4a: $\Gamma_{1,\mathrm{local}} = \sum z_i C_i$ (gh 0);
- ST4b: $P_{\mathrm{ret}}(S, C) = -P_{\mathrm{ret}} A_1$ (gh 1);
- retained identity: $(S, \Gamma_1) = \mathcal{A}_1$ at one loop and order κ^2 in the retained sector.

Rows. All 23 one- and two-field monomials of ghost number 0 and 1 are classified, each with its own justification. Two independent one-loop enumerations agree with XVI-P: $E = 2$ gives 9 graphs and $E = 1$ gives 3.

Counterterms (ST4a). The MS-bar counterterm, $-$pole, in the 11-direction declared p14r domain over $\mathbb{Q}(i)$:

- $h{\cdot}h$: $\frac{49i}{480}, \frac{41i}{160}, -\frac{49i}{240}, -\frac{41i}{80}, \frac{43i}{120}$;
- $c{\cdot}\bar{c}$: $-\frac{3}{8}, \frac{1}{4}$, in the route-G orientation; these equal the D06 poles up to the orientation sign -1;
- $c{\cdot}K'$: $0, -\frac{3}{8}, \frac{1}{4}$;
- $\phi{\cdot}\phi$: 0. P19 carried this coordinate as a placeholder; it is now earned.

The p22 evanescent counterterms ($K'c$ E2: $-\frac{1}{16}$; $\bar{c}c$ SE: $\frac{1}{8}$; hh sum) subtract the evanescent poles only. They lie outside the p14r 4D domain as a disclosed contract revision. They are not needed for the identity: the ch defect is zero with $x = 0$.

ST4b and closure.

- $c \cdot h$: the defect is zero exactly in d (P19).
- $c \cdot b$: this is the ghost equation, and its Σ equals D06.
- $c \cdot \bar{c}^*$: unreachable.
- $c \cdot \phi$: zero by parity.

Full σ_0 closure of the complete MS-bar counterterm holds in every gh+1 sector. The σ_0 rank on $\phi{\cdot}\phi$ is 0, and the cohomology of the domain is 0. **Outcome: cancellable, with no residual.**

13 O.1 walk (verbatim ledger)

1. *State retained quantum identity*: earned. XVI-O status was Constructed_as_statement.
2. *List amplitudes constrained by identity*: earned. XVI-O status was "manually_specified_partially_checked". Completeness now rests on two mechanical enumerations.
3. $S^{[2]}_{\mathrm{inv}}$ *classical consistency*: earned in Q01.
4. *Evaluate inventory graphs*: earned. All 12 graphs have exact values.

5. *Remainder cancellation / CT solution*: earned, as above.

6. *Enlarge target*: not triggered. At one loop the κ^2 part of (S, Γ_1) is $(S_0, \Gamma_1[\kappa^2]) + (S_1, \Gamma_1[\kappa])$. $\Gamma_1[\kappa]$ is one-point only and is 0, and Γ_1 with three fields starts at κ^3.

All eight closure-check reasons are answered. Enlarge rule 3 is respected: no XVI-M graph is used or identified with an action graph. Downgrade rule 2 is respected: no amplitude is omitted without its own justification.

O1_closure = CLOSED_PENDING_INDEPENDENT_REVIEW (confirmed: false). The scope is the one-loop κ^2 bulk two-point retained sector of S_{inv} at $m = 0$, with the frozen gauge fermion, under the declared prescription, with external momenta in QdS. The functional ST4b (all n-point) and bulk symmetry restoration are *not* claimed.

14 Guard, gate and records

- The guard refuses any `Poly` that has d as a generator. d inside the coefficient domain is admitted, as used by the XVI-P integral library. `apart` and `apart_list` are removed from the allow-list. 18 bypass forms $\times$ 4 expressions are probed, and all of them raise.

- The gate compares every Q01/Q02 certificate value path by path against `q02_expected.json`, whose sha256 is pinned in the gate. It independently checks the O.2(c) cells (exact set, all true, $\kappa^0..\kappa^2$, including the build input), each per-file pin, the conventions files and the errata file, and fails closed.

- `tools/build.py` no longer syncs by default; a release opts in with `--sync`.

- The registration record, errata E2 (severity low) and E4 (unpinned `p23_conventions`) are corrected, and the XVI-N label note (r10 file, r12 content) is added.

Q01 layer

15 XVI-Q Q01: scope and registration

XVI-Q is a new landmark (registered locally: `registration/local_registration.json`; not published). Its aim is to import XVI-P's verified ℓ_4 to discharge O.2, to state the bulk identity, and to attempt O.1 closure. XVI-O, XVI-N and XVI-P are read-only. They are imported by sha256 (`intake/imports.json`; XVI-P frozen at p23, source zip `3f4a88db...`). Q01 covers the intake and O.2. O.1 stays NOT_CLOSED, bulk symmetry is not claimed, and the insertion stays frozen.

O.1 target (XVI-O ledger, verbatim). "one-loop_order_kappa2_bulk_two_point_on_S_inv": bulk two-point (graviton, scalar, ghost), including the Einstein–scalar hh channels and the vertex-derived gh= 0 BV/source tree seeds; baseline $S_{\rm inv}$; frozen gauge fermion; $m=0$. Closure requires a *verified* retained quantum identity with O.2, O.4, O.5, O.6 and O.7 earned.

Errata from the p23 review (intake).

- E1 (medium): the p23 O.5 line "all $E=2$ graphs evaluated" overclaims. D04 and D05a were never evaluated, D06 only as the ghost self-energy, and D12 is Missing. Bulk O.5 is open.
- E2: the stated reason "grav ℓ_4 Incomplete" is stale.
- E3: guard bypasses (fixed here).
- E4: gate value gaps (fixed here: every number is value-checked).
- E5: scaleless 5-leg tadpoles are missing from the κ^3 inventory.

16 O.2: the classical κ^2 master $2(S_0,S_2)+(S_1,S_1)=0$ on $S_{\rm inv}$

(a) Full ℓ_4 — Constructed. XVI-P's p23 stages, re-executed from the pinned zip, reproduce the $\ell_{2,3,4}$ exports byte-identically. The ℓ_4 export (3360 terms) equals, term by term, an *independent* reconstruction from the $\Gamma\Gamma$ Lagrangian $\sqrt{-g}\,g^{mn}(\Gamma^a_{mb}\Gamma^b_{na}-\Gamma^a_{mn}\Gamma^b_{ab})$. That reconstruction uses exact ϵ-grading, g^{-1} and $\sqrt{-g}$ to second order in H, and a sparse ring over $\mathbb{Q}$. Its only normalisation, -2, is fixed by XVI-O's conformal probe and coincides with XVI-O's convention $\ell_4=-2[\epsilon^4](\sqrt{-q}R)$, since $L_{\Gamma\Gamma}=\sqrt{-g}R+\partial B$. A one-term mutation is detected.

(b) Explicit $O(\kappa^2)$ antibracket expansion — Constructed. With $S_{\rm EH}=\sum_n \kappa^{n-2}\int \ell_n$, $s=s_0+\kappa s_1$ ($s_2=0$, derived), and $S_{af}\ni\int h^{*mn}(sh)_{mn}$, the field projection is

$$(S,S)|_{af0,\rm grav}=2\int\sum_{m\le n}\frac{\delta S_{\rm EH}}{\delta h_{mn}}(sh)_{mn}.$$

Its κ^2 coefficient is $2\int W_2$ with $W_2=\sum E_{mn}(\ell_4)(s_0h)_{mn}+E_{mn}(\ell_3)(s_1h)_{mn}$. This is the component form of $2(S_0,S_2)+(S_1,S_1)$. W_2 is linear in c, so $\int W_2=0$ for all c iff $E_{c^\mu}(W_2)=0$. On an independent XVI-Q jet engine, W_2 has 15428 terms and $E_{c^\mu}(W_2)=0$ exactly for $\mu=0..3$; the κ^0 and κ^1 regressions also vanish. Doubling one ℓ_4 term, or dropping s_1, breaks the result.

(c) Ghost/antifield sector — Constructed (re-executed evidence). In XVI-P's exact P13 master (re-executed; content-identical to p23), every κ^2 cell vanishes: af0 ghost/gauge fixing (transformed), and af1 and af2 in both invariant and Ψ-transformed variables. s_2 is *derived* zero in all sectors, from the affine g, the linear Lie derivative, and the ghost law; it is not invented. This answers XVI-O's "absent under inherited s". s is nilpotent exactly. Seven mutations each produce residuals. Caveat: Q01 has no second, independent engine for this sector.

Outcome. O.2 at κ^2 is **Constructed in XVI-Q**; XVI-O's own record is unchanged (read-only). The scope is classical, $S_{\rm inv}$, $m=0$: not quantum, not the prepared S_D, not the insertion.

17 d guard (carried)

The XVI-P p23 dred is imported unchanged, with one added rule. Polynomial construction (`Poly.new`, a class attribute) and any internal rename of d are admitted only when the *entry* sympy function called from user code is an allow-listed rational normaliser; otherwise they are denied. This closes the p23 bypasses: `sympy.polys.rem`, pre-import capture, `reduced`, `groebner`, the Dummy rename and `rep.eval`. Probe: 16 forms $\times$ 4 expressions all raise, and ordinary algebra is unchanged.

18 Status and Q02 plan

After Q01: O.2 Constructed; O.4 inventory (with E1); O.5 bulk open; O.6/O.7 bulk open; enlarge rule not triggered at κ^2. Hence O.1 is NOT_CLOSED. Q02 will:

1. evaluate the bulk O.5 graphs D04, D05a and D06 (full $c\bar{c}$ two-point), and settle D12 (Missing, never zero);
2. state and verify the bulk retained identity ST4a/ST4b on the full two-point sector;
3. solve for the bulk counterterms and attempt O.1.